84

98 96%

94%

Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

FORMATIVE ASSESSMENT

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About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Sesa Mamoloko	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: Sesa Mamoloko S Sesa M.G

Learners Name: Sesa Mamoloko

Learner's signature: S Sesa M.G Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____

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Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Sesa Mamoloko	Signature Sesa.M.G
Facilitator Name:	Signature:

TASTC
TECHNOCENTRIC
VARSITY

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LEARNING UNIT 1:

KM-07-KT01: Concepts, theories and principles of supply Systems (20%)

Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

Task	Questions Description	Marks
1.	a) Explain the generation and differences between single and three phase alternating current by using wave forms and vector diagrams.	4
	b) Explain by drawing wave form diagrams, the differences between line and phase values.	4
	c) Explain the generation principles of direct current.	4
	d) Calculate power and energy in direct and alternating current circuits	4
	e) Explain with the aid of drawings the operating principles of direct current generators.	4

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$P = VI \cos\theta$

- Includes power factor ($\cos\theta$)

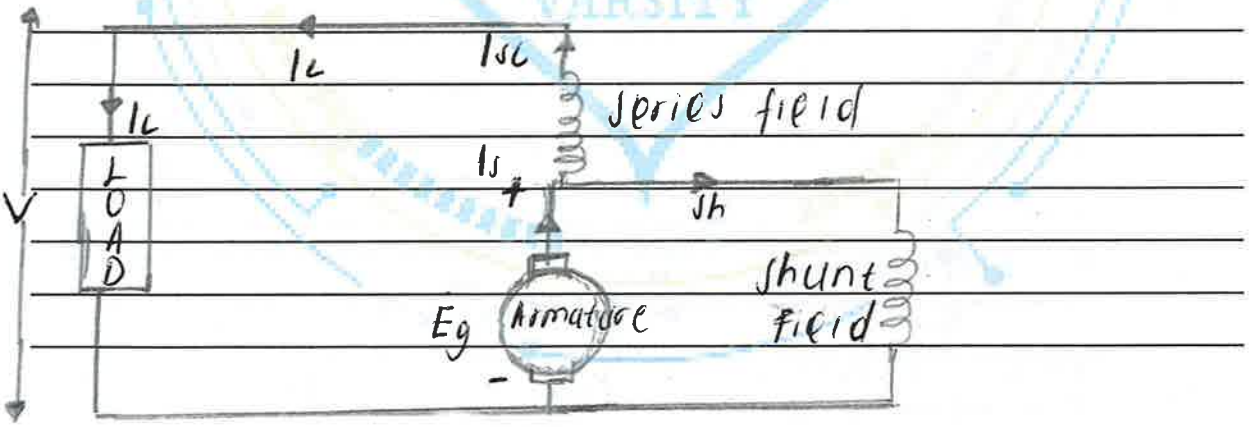
- Apparent power $S = VI$ (VA)
- Reactive power $Q = VI \sin\theta$ (VAR)
- Energy $E = P \times t$ (measured in KWh)

(E) Basic parts :

- Armature (rotating coil)
- magnetic field
- Commutator
- Brushes

Operation

- Armature rotates in magnetic field
- EMF is induced (AC internally)
- Commutator converts it to DC.
- Brushes collect current to external circuits.



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(B)(i) Primary cells (Non-rechargeable)

* Advantages
- Ready to use (no charging required)
- Long shelf life (low self discharge)

* Disadvantages
- Cannot be recharged
- Environmental waste

(ii) Secondary cells

* Advantages
- Rechargeable
- Lower long term cost.

* Disadvantages.
- Higher initial cost
- Require charging equipments.

(D)

Lead-Acid Battery Charging.



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LEARNING UNIT 3: KM-07-KT03: Transformers (20%)	
Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

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(B) mutual inductance occurs when a 1-ampere change in the primary current induces 1-volt in the secondary coil.

(C) ① Core Losses (Iron loss)

- Hysteresis Loss

* Cause: caused by the friction of magnetic domain, in the core as they flip back and Fourth with the alternating magnetic Field.
* Effect: Energy is depleted.

② Copper losses

- Due to resistance in windings (I^2R losses)

Effects: Heating and Reduced efficiency.

(D) Transformer ratios.

- Voltage ratio $\frac{V_1}{V_2} = \frac{H_1}{H_2}$
- Current ratio $\frac{I_1}{I_2} = \frac{H_2}{H_1}$

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LEARNING UNIT 4:

KM-07-KT04: Types of cables and applications (20%)

Learning Outcome 1:	KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.
Learning Outcome 2:	KT0402 Cable materials and their functions and characteristics
Learning Outcome 3:	KT0403 Identification of cable characteristics and properties
Learning Outcome 4:	KT0404 Applications of various types of cable
Learning Outcome 5:	KT0405 Installation methods and safe use of cables
Learning Outcome 6:	KT0406 Safe transport and storage of cables

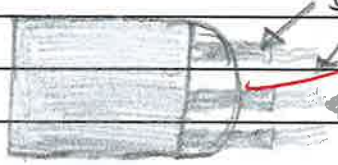
Task	Questions Description	Marks
4	a) Describe, with the aid of sketches, the different types of cables and their construction, and explain the function of the various materials used	4
	b) Describe the factors effecting the effeciency of cables	3
	c) Describe the precautions required when transporting and storing cables	3

(A)

Armouring ← Sheath

Bedding insul | main metallic

outer sheath



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LEARNING UNIT 5:

KM-07-KT05: Switchgear and control gear (20%)

Learning Outcome 1:	KT0501 Principles of operation of switchgear and control gear
Learning Outcome 2:	KT0502 Components of switchgear and control gear systems and the application thereof.
Learning Outcome 3:	KT0503 Electrical drawings
Learning Outcome 4:	KT0504 Introduction to Programmable Logic Controllers (PLC's)
Learning Outcome 5:	KT0505 Introduction to Soft starters
Learning Outcome 6:	KT0506 Introduction to Variable speed drives

Task	Questions Description	Marks
5	a) Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and operating mechanisms	4
	b) Describe the basic principles of operation for programmable logic controllers	3
	c) Describe the basic principles of operation for variable speed drives	3

a, Disconnectors (Isolators)

i, Construction : > Fixed and moving contacts
> Insulated base
> Operating handles

ii, Operation : When current flows in coil, magnetic field is created.

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B, A PLC - is a digital controller used to automate processes

* Basic Operation: 3

① Input stage - Recives signal from sensors / switches
② Processing stage - Cpu proleses according to program
③ Output stage - Sends signals to devices (Motors, lamps,)

Key features:
> Programmable
> Reliable
> Used in Industrial Automation.

C, A varriable speed drive → Controls motors speed by changing supply frequency

* Working principle:
① AC supply - converted to DC (Rectifier)
② DC → converted back to AC (Inverted)
③ Output frequency is varied → controls motor speed

* Advantages: Energising saving
: Smooth starting
: Speed control

3

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• Frame and shaft
> mechanical support and rotation

(B) Single phase motors
- Not self starting
- Low efficiency

• Three phase motor
- Self starting
- High efficiency

(C) (1) No load Test
> Motor runs without load
> measures losses

(2) Load Test
> Motor runs with load
> checks performance

(d) (1) power in Ac Motor : $P = V \times I \cos\theta$
$P = VI \cos\theta$

(2) Motor efficiency $\eta = \frac{\text{output power}}{\text{input power}} \times 100\%$

(3) Speed of Ac Motor $N_s = \frac{120F}{P}$

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ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS
Good work

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met** ☒	**Requirements not met** ☐
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS
Good !!

DECLARATION BY THE FACILITATOR
I ..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

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Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

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3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments 4. Conduct observation in workplace where applicable 5. Complete learner toolkit 6. Complete assessor toolkit

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